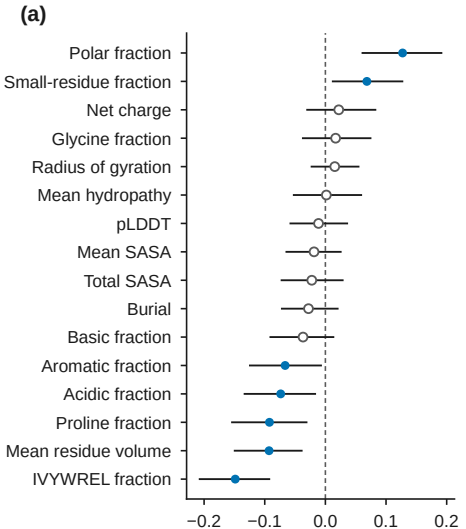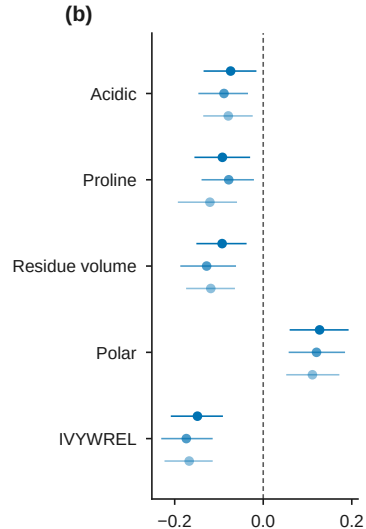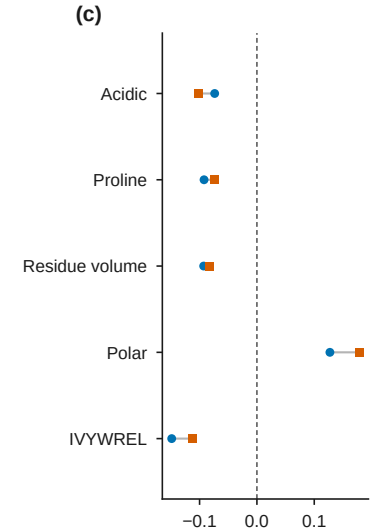Thermal adaptation is absent from the catalytic neighbourhood